Bitcoin Price Prediction and Trend Analysis

Overview

This repository contains the code and resources for the Bitcoin Price Prediction and Trend Analysis using Transformer Models and RSI-Based Insights thesis project. The research explores deep learning techniques, particularly Transformer-based models, to forecast Bitcoin prices and analyze market trends.

Features

- Transformer-based Time Series Forecasting (PatchTST)
- Comparison with ARIMA, Prophet, LSTM & Variants, XGBoost
- RSI and MACD Integration for improved forecasting
- Backtesting on historical Bitcoin price data
- Evaluation using multiple metrics (MSE, RMSE, MAE, Accuracy, F1-Score, R^2)

Installation

1. Clone the repository:

   git clone https://github.com/shakibaMB/Bitcoin-Price-Prediction.git

   cd Bitcoin-Price-Prediction

2. Install dependencies:

   pip install -r requirements.txt

3. Run the model:

   python main.py

Results

The study found that PatchTST consistently outperformed other models across different forecasting windows, achieving higher accuracy and lower error rates. LSTM models performed well but struggled with longer time horizons, while traditional models like ARIMA and Prophet were less effective due to Bitcoins high volatility.

Future Work

- Incorporate sentiment analysis from news and social media.
- Extend to other cryptocurrencies (Ethereum, Litecoin, etc.).
- Develop a real-time forecasting API for live predictions.
- Improve model interpretability using SHAP and attention visualization.

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